

Supplementary Material

This supplementary material lists implementation details used in the GrapeMaster platform. The main manuscript reports the platform architecture, data flow, analytical modules, validation design, and workflow verification; this document provides supplementary workflow figures, the software stack, selected data-model attributes, and backend-record screening rules for reproducibility and technical inspection.

Supplementary figures

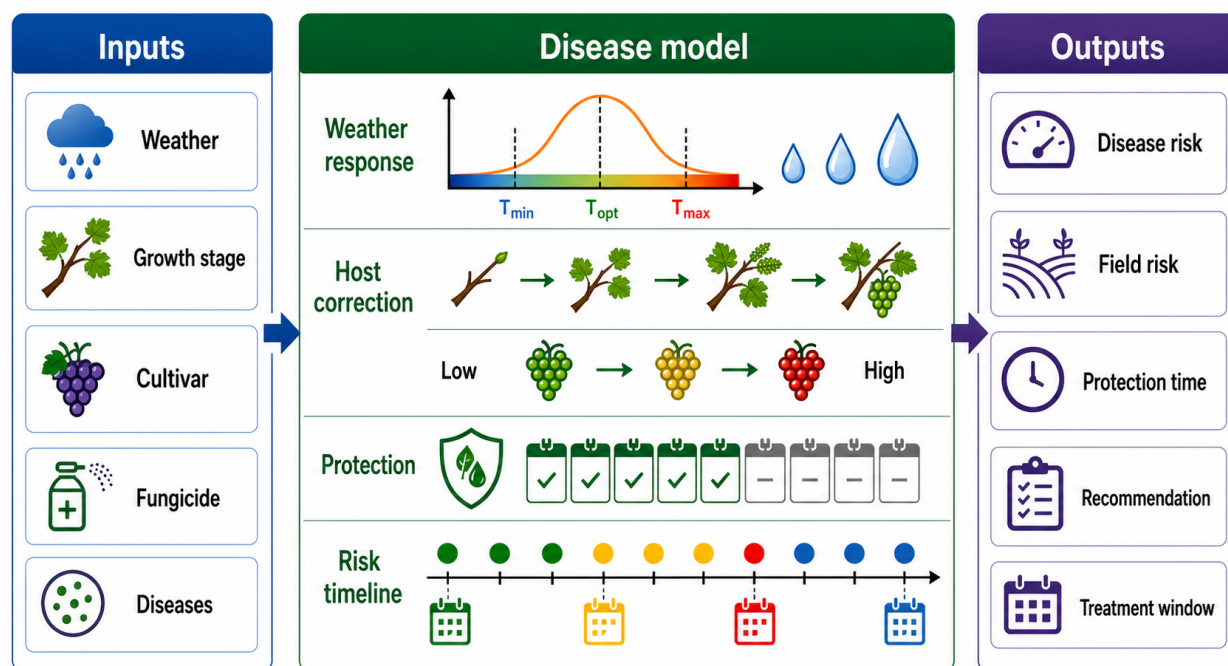


Figure S1. Platform-compatible structure of the GrapeMaster disease-risk engine. Weather, growth-stage, cultivar, protection-status, and target-disease inputs are converted into disease-risk states, field-risk states, protection timing, recommendations, and treatment-window outputs through the platform workflow.

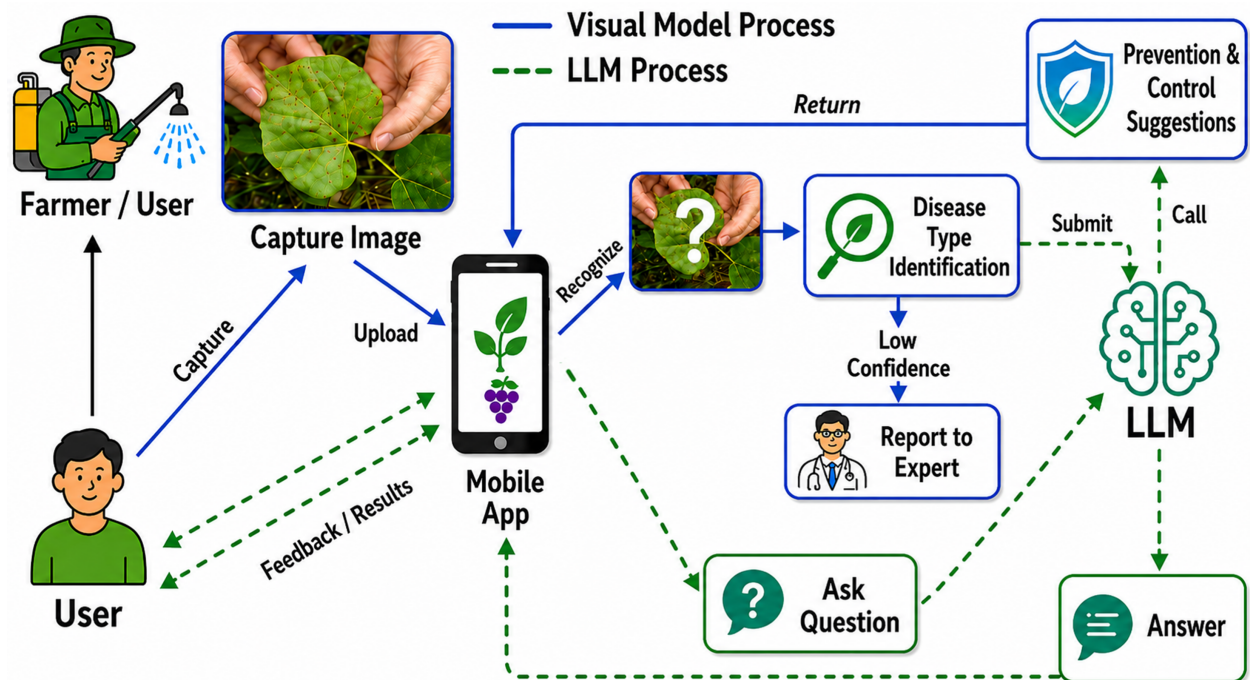
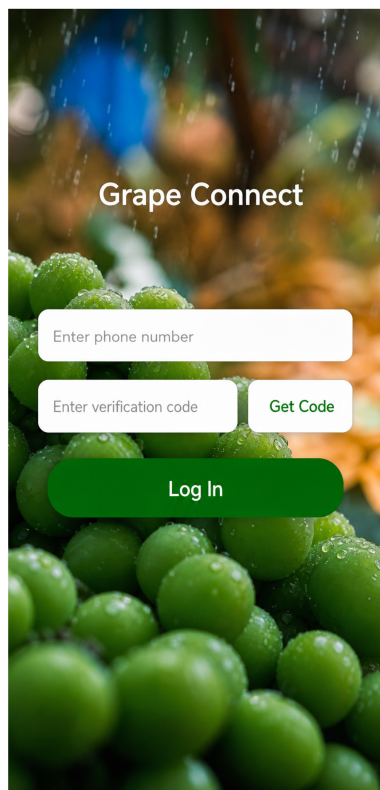
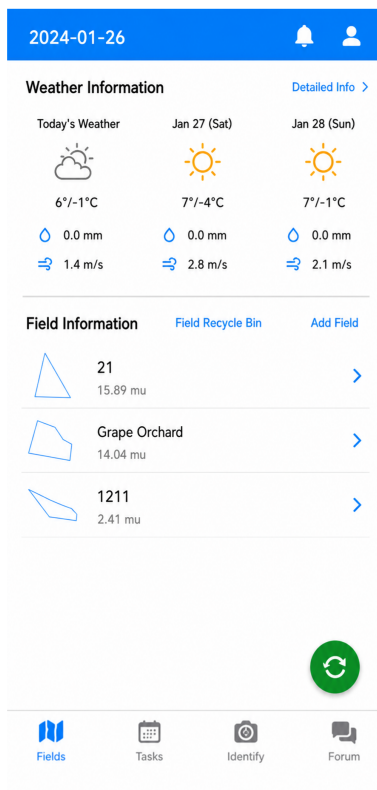


Figure S2. Image-recognition and advisory workflow in GrapeMaster. Field images are uploaded through the mobile application, screened by the visual model, and returned as candidate disease categories with confidence information. Low-confidence cases can be routed to expert review, while the LLM channel provides auxiliary agronomic consultation based on the recognition context.



(a)



(b)



(c)

Figure S3. Mobile field-management interfaces in GrapeMaster. The field workspace provides the operational entry point for field creation, field inspection, crop-season access, field sharing, deletion, recovery, and linked management functions.

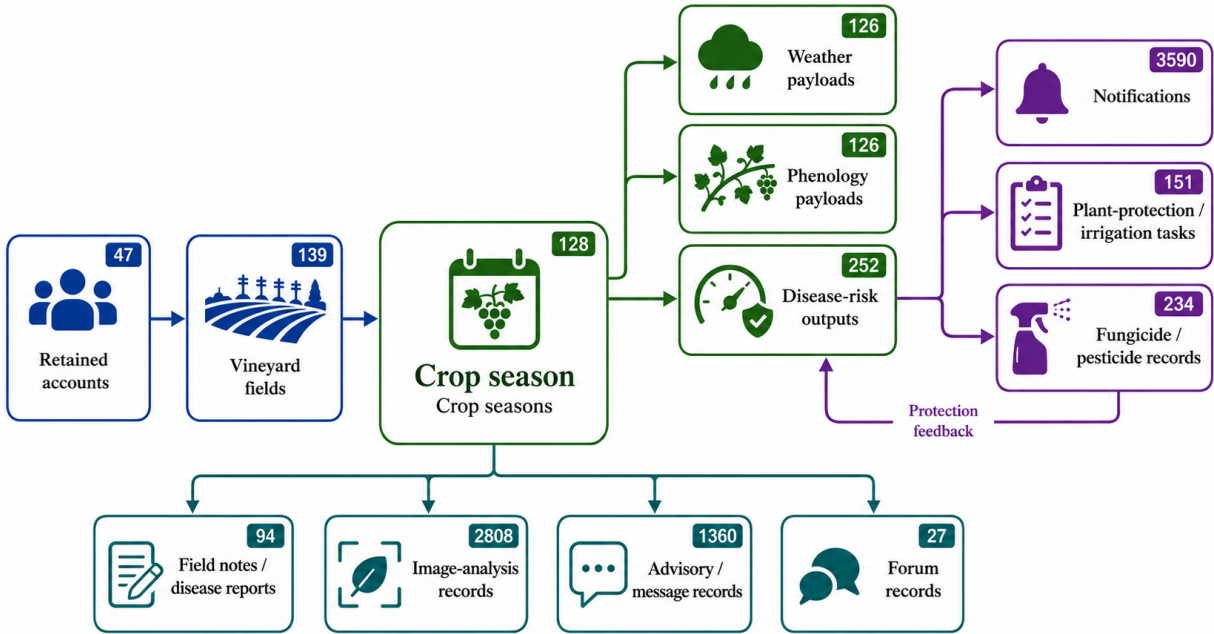


Figure S4. Backend linkage topology among retained GrapeMaster records. The crop-season record is shown as the central anchor connecting account-field setup, analytical payloads, risk outputs, notifications, tasks, protection feedback, image-analysis records, advisory messages, and forum records. Numbers in the corner labels indicate retained record counts for each record class in the screened backend export.

Supplementary tables

Table S1. Additional module-level technical details supporting the platform-compatible analytical components.

Module detail	Main implementation or evaluation element	Role in the main manuscript
Phenology calibration	Guangxi phenology observations were mapped to broad BBCH-related major stages and evaluated by grouped site-year validation using exact agreement, user-visible agreement, adjacent-stage agreement, severe-mismatch rate, and mean absolute stage distance.	Supports the crop-stage context used for mobile display, risk interpretation, and task timing.

Module detail	Main implementation or evaluation element	Role in the main manuscript
FSIM-S disease-risk module	The downy mildew module used platform-transformed weather, crop-stage sequence, cultivar susceptibility, target-disease information, and fungicide-history input to produce date-aligned disease-risk states, protection states, recommendations, and treatment windows.	Supports first seasonal infection-period warning and protection-state interpretation within the traceability workflow.
FSIM-S spore-pressure term	Field-derived sporangia pressure was used in the evaluated configuration to adjust the infection-triggering threshold; this is a model-level enhancement and an optional sensor-enhanced pathway.	Provides module-level evidence for Guangxi-oriented risk modelling but is not required as a live input for routine platform operation.
Image-recognition model selection	Candidate visual models were evaluated with grape-focused validation metrics, including accuracy, precision, recall, and F1-score; the selected classifier provides disease labels and confidence values for the advisory interface.	Supports post-symptom image evidence within the crop-season record context.
LLM-assisted advisory	Recognition outputs are converted into disease-specific consultation context before advisory text is generated; advisory messages are stored or linked to the crop-season context where identifiers are available.	Provides auxiliary explanation and consultation records; it does not replace validated models, expert diagnosis, pesticide labels, or local regulations.

Table S2. Formal crop-season traceability object used in the GrapeMaster workflow.

Object element	Included records or payloads	Traceability function
Season identity	Crop-season identifier, linked field identifier, crop year or season context, cultivar, cultivation method, vine age, field coordinates, and field size	Defines the relational and temporal anchor for one vineyard management season

Object element	Included records or payloads	Traceability function
Environmental state	Weather records, historical and forecast weather payloads, transformed hourly and daily weather inputs	Preserves the environmental exposure used by phenology, disease-risk, and operation-suitability workflows
Crop-development state	Phenology payloads, broad growth-stage labels, BBCH-related stage information, and stage descriptions	Links crop-stage interpretation to warning windows, advisory timing, and field-task planning
Disease-risk state	Disease-risk payloads, request-response bodies, field-risk states, protection states, treatment windows, and action recommendations	Connects model input, model output, and user-facing warning interpretation under the same seasonal anchor
Operation state	Notification records, plant-protection tasks, irrigation tasks, task status, execution date, and overdue state	Tracks whether risk or management events were delivered and converted into executable field operations
Treatment feedback	Fungicide or pesticide records linked to plant-protection tasks, product information, dosage information, and application date	Converts completed field operations into evidence for later protection-state interpretation and risk recalculation
Symptom and advisory evidence	Image-analysis records, advisory or message records, disease reports, field notes, and forum records where linked to account, field, crop season, or image identifiers	Preserves symptom evidence and season-long advisory interaction for later interpretation and review
Replay archive	Shared crop-season keys, task keys, payload availability, linkage checks, explicit absence notes for optional symptom records, and representative replay summaries	Allows the season to be reconstructed as a prediction-action-feedback-review chain

Table S3. Database relationship rules supporting crop-season-centered traceability.

Relationship rule	Required linkage	Verification purpose
Account-to-field linkage	Retained account identifiers must link to at least one vineyard field record	Confirms that the account has an operational vineyard context rather than registration-only activity
Field-to-crop-season linkage	Each evaluated crop-season record must be linked to a field record through field or crop-season identifiers	Separates persistent field identity from season-specific analytical and management state
Crop-season-to-analytical-payload linkage	Weather, phenology, disease-risk, and model request-response payloads should share the crop-season identifier where applicable	Confirms that analytical inputs and outputs can be retrieved under the same seasonal anchor
Crop-season-to-event linkage	Notifications, warning events, and task records should be associated with the retained user, field, or crop-season context	Confirms that analytical or management events can be connected to the operational season
Task-to-product linkage	Fungicide or pesticide product records must link to plant-protection tasks through task identifiers	Confirms that completed operations can be converted into product-level treatment feedback
Crop-season-to-evidence linkage	Image records, advisory records, field notes, and disease reports should retain available account, field, crop-season, image, or message identifiers	Confirms whether symptom evidence and advisory interactions can be interpreted together with seasonal state
Explicit absence rule	If an expected evidence type is absent for a replay anchor, the absence is recorded rather than inferred as failure of unrelated workflow steps	Prevents overclaiming and separates missing symptom evidence from successful risk-task-feedback linkage

Table S4. Workflow state-transition rules used to interpret the risk-task-protection-feedback chain.

State transition	Trigger or record basis	Interpretation
Field setup to crop-season setup	A field record is created and a crop-season record is attached to that field	Establishes the seasonal management object
Crop-season setup to state updating	Weather and phenology payloads are generated or attached to the crop season	Provides environmental and crop-development state for risk interpretation
State updating to risk interpretation	Disease-risk request-response payloads are attached to the same crop season	Converts weather, phenology, and management inputs into date-aligned risk states
Risk interpretation to event delivery	Warning or management notifications are generated under the retained account or crop-season context	Documents that analytical state was translated into user-facing events
Event delivery to task creation	A plant-protection or irrigation task is created from or within the crop-season workflow	Converts a warning or management context into an executable operation record
Task creation to task completion or non-completion	Task status, planned date, execution date, completion state, or overdue state is stored	Distinguishes pending, completed, and missed operations
Task completion to treatment feedback	Completed plant-protection tasks are linked to fungicide or pesticide product records	Converts the field operation into product-level management feedback
Treatment feedback to renewed interpretation	Product and execution records are available for protection-state interpretation or later risk recalculation	Allows later risk states to be interpreted in light of recent management history
Season replay	Shared crop-season and task identifiers retrieve the sequence of state updating, risk interpretation, event delivery, task execution, feedback, and optional symptom/advisory record availability	Reconstructs the management chain for audit and later evaluation

Table S5. Workflow verification rules used for backend-record and representative-replay checks.

Verification check	Operational criterion	Supported interpretation
Operational-account retention	An account is retained only when it has business records and field-crop-season linkage after deduplication and target-context screening	Defines the evaluated operational account set
Analytical payload completeness	A crop season is counted as analytically complete when weather, phenology, disease-risk, and request-response payloads are present under the shared seasonal context	Shows whether the analytical state can be reconstructed for that season
Risk-to-operation linkage	Notification and task records are checked against retained user, field, or crop-season context	Shows whether risk or management events entered the operational workflow
Treatment-feedback linkage	Fungicide or pesticide records are checked through task identifiers and task completion status	Shows whether field operations can be converted into feedback records
Evidence-linkage availability	Image-analysis, advisory, disease-report, and note records are checked for available account, field, crop-season, image, or message links	Shows whether symptom evidence and advisory interactions are available for seasonal interpretation
Representative replay	A selected crop-season anchor must retrieve weather, phenology, risk, notification, task, and fungicide records through shared crop-season or task keys	Demonstrates reconstructability of one risk-to-feedback chain
Boundary reporting	Missing optional symptom or advisory records under the replay anchor are explicitly reported rather than omitted	Separates verified linkage from unsupported production-outcome claims
Excluded outcome claim	Backend linkage and replay checks are not treated as evidence of pesticide reduction, disease-control efficacy, economic benefit, or user adoption	Maintains the distinction between platform-readiness evidence and agronomic-impact evidence

Table S6. Software stack and implementation components used in the GrapeMaster platform.

Layer	Component group	Main technologies or libraries
Mobile client	Application framework	Dart, Flutter
Mobile client	Android shell and build configuration	Kotlin, Gradle
Backend business service	Web framework and API layer	Python, Django 4.2, Django REST Framework
Backend business service	Authentication, scheduling, and asynchronous communication	SimpleJWT, Django APScheduler, Django Channels
Backend business service	Database access and persistence	PostgreSQL, psycopg2, normalized relational tables, JSON model payloads
Disease-model service	Service interface	FastAPI, Uvicorn, Pydantic, Starlette
Weather and time-series processing	Numerical and date-time processing	NumPy, Pandas, Python Dateutil, Pytz
Weather and geospatial processing	Field geometry, geocoding, and weather-service access	Shapely, Fiona, GeoPandas, Geopy, Requests, HTTPX, Open-Meteo client packages, request caching, retry utilities
Phenology and risk analytics	Crop-stage and risk-state calculation	Python analytical workflows, date-aligned weather transformation, cultivar and crop-season metadata
Image processing and recognition	Image preprocessing and deep-learning inference	PyTorch, TorchVision, OpenCV, Pillow, Ultralytics-related components, EfficientViT-related modules, segmentation components
Advisory interface	Text advisory channel	ChatGLM3-derived advisory workflow following the published CDIP-ChatGLM3 framework

Table S7. Selected data-model attributes supporting the crop-season-centered workflow.

Record group	Main attributes or stored content	Platform role
Field record	User linkage, field name, area, GeoJSON boundary, centroid coordinates, administrative region, sharing state, deletion state, notice state, and generated field-boundary preview	Persistent spatial and ownership unit for vineyard management
Field geometry processing	Boundary parsing, boundary-image generation, area estimation, centroid calculation, and optional geocoding request	Converts user-defined field boundaries into location and display objects
Crop-season record	Field linkage, grape cultivar, cultivation method, vine age, field coordinates, field size, and cultivar metadata	Primary seasonal anchor for weather, phenology, risk, tasks, products, images, advisory records, and summaries
Master-data records	Grape variety, cultivar susceptibility, cultivation method, BBCH stage, disease type, fungicide, pesticide, and reminder rules	Controlled reference values for field, crop-season, task, risk, and advisory records
Model payload records	Weather, phenology, disease-risk, and request-response payloads stored with structured records	Preserves inputs and outputs for mobile display, recalculation, and audit

Table S8. Backend-record screening rules used before summarizing deployed GrapeMaster records.

Screening step	Rule	Rationale
User-record integration	User records were integrated using available user identifier, custom-user identifier, and phone-number fields.	Avoids duplicate counting of account-level records

Screening step	Rule	Rationale
Business-record filter	Accounts with no business records across field, crop-season, notification, task, fungicide or pesticide, advisory or message, forum, disease-report, and note tables were excluded.	Removes registration-only accounts from operational summaries
Field-crop-season linkage filter	Accounts without both field records and crop-season records were excluded.	Retains records that can support crop-season-centered workflow evaluation
Target-context filter	A single account associated only with Linwei District was excluded.	Removes a non-target-region account from the Guangxi vineyard evaluation context
Retained account set	Forty-seven accounts remained from 95 deduplicated accounts.	Defines the candidate account set used for operational-record coverage